



Airline Baggage Complaints

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01

Overview



Airline Baggage Complaint Dataset

- This dataset contains **252** records of monthly baggage complaints to **American Eagle, Hawaiian, and United** from **January 2004 to December 2010**. Each record lists:
 - Airline
 - Date, Month, Year
 - # of baggage complaints
 - # of flights scheduled
 - # of flights cancelled
 - # of passengers who boarded a plane

	Airline	Date	Month	Year	Baggage	Scheduled	Cancelled	Enplaned
0	American Eagle	01/2004	1	2004	12502	38276	2481	992360



Objective

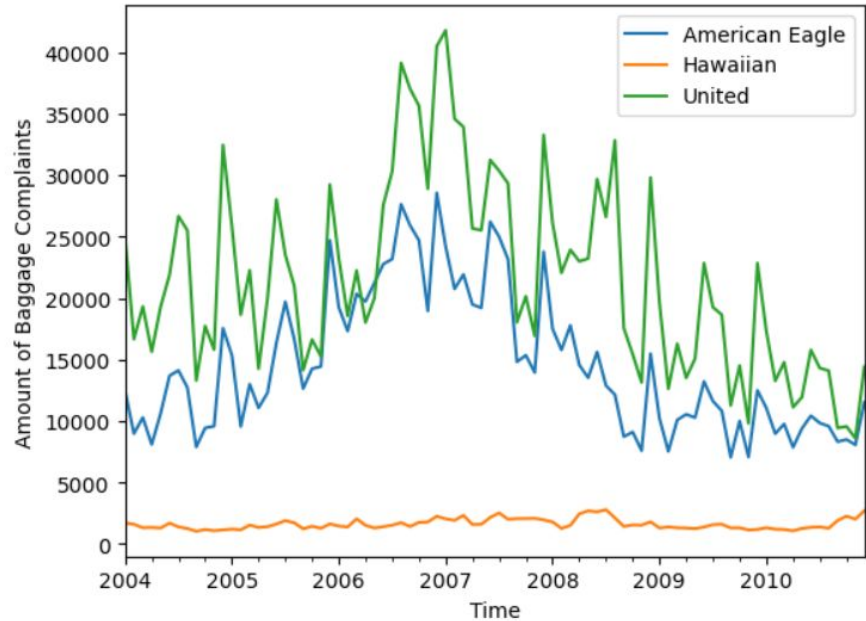
We plan to use the features of this dataset to predict the corresponding number of baggage complaints. These predictions could be useful for a company to estimate their staffing needs during different periods in the year.



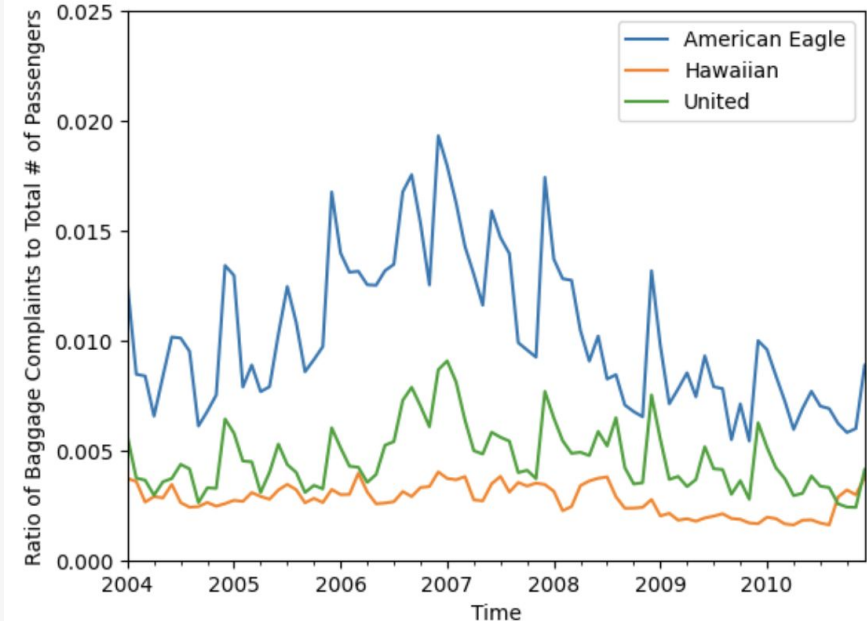
02

Exploratory Data Analysis

of Baggage Complaints vs Time

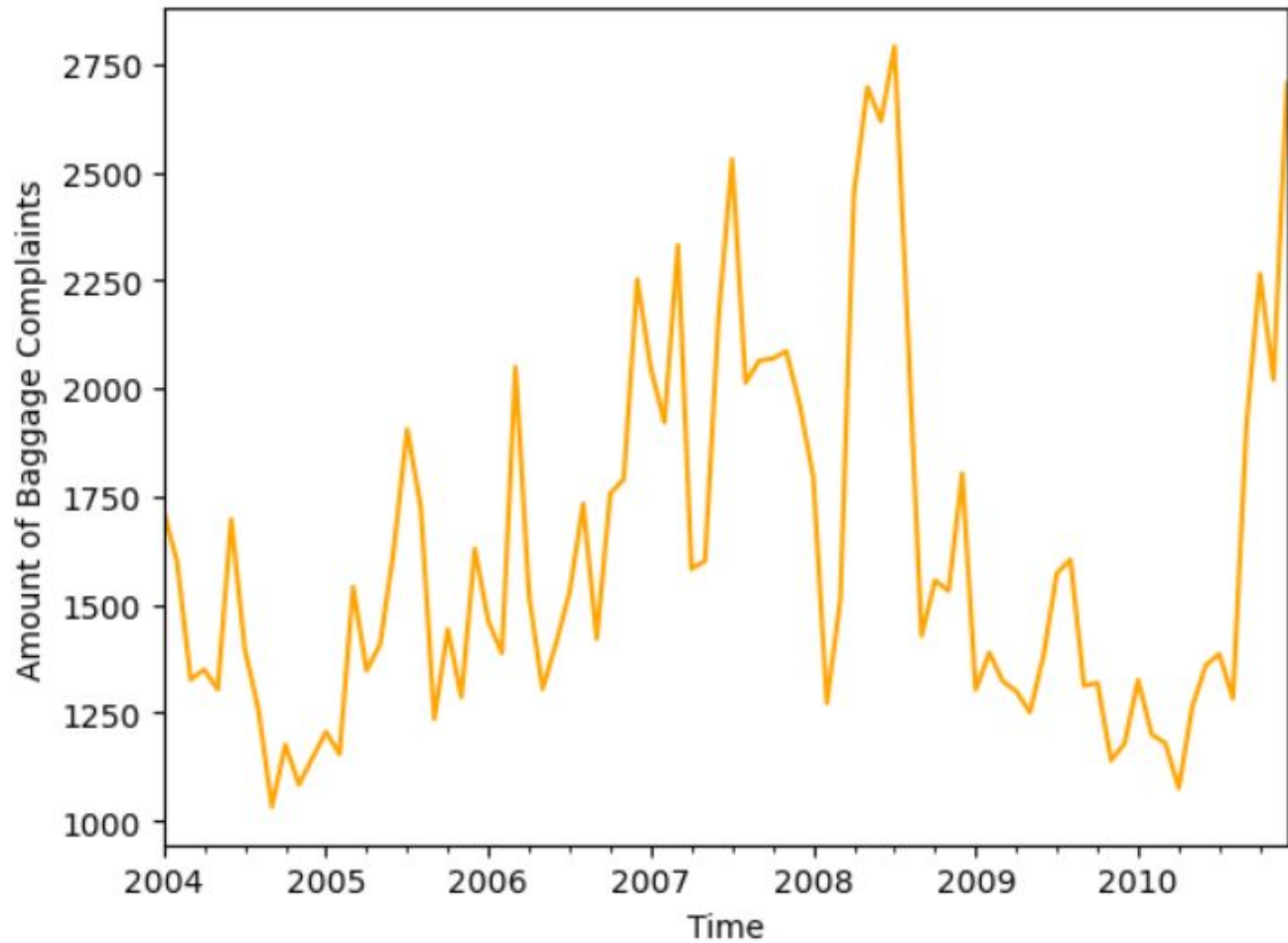


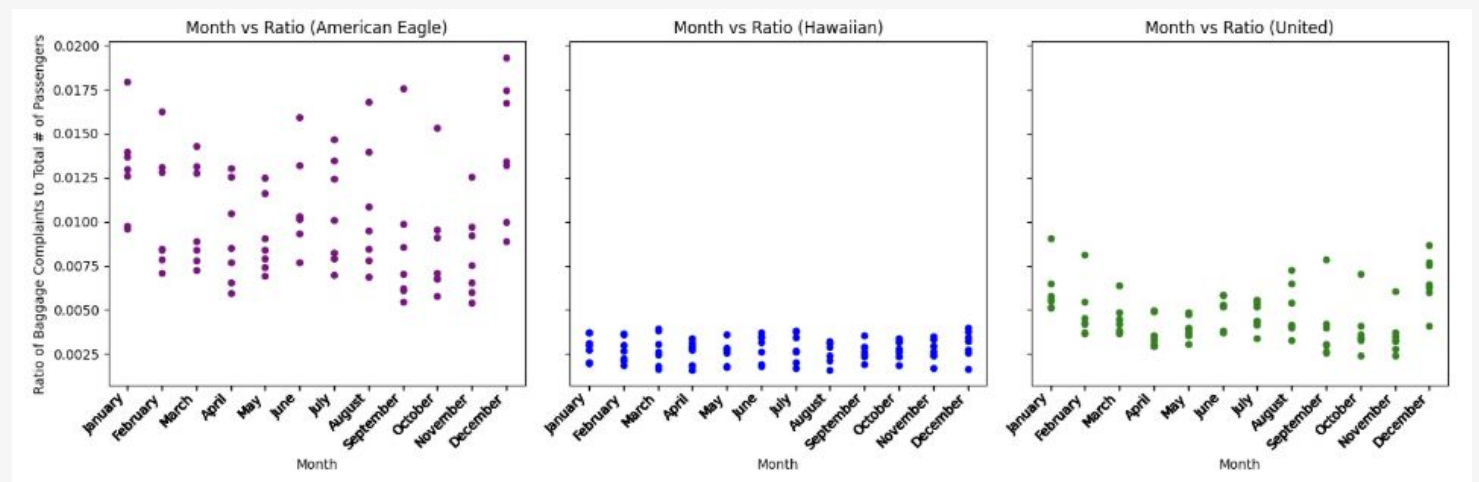
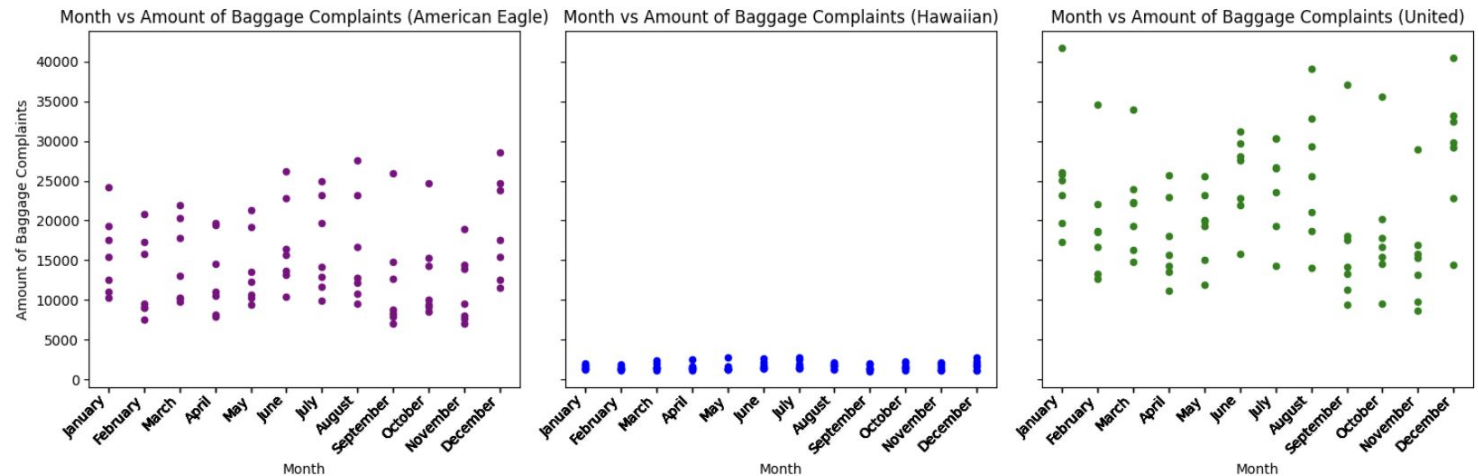
Ratio(Baggage/Passengers) vs Time



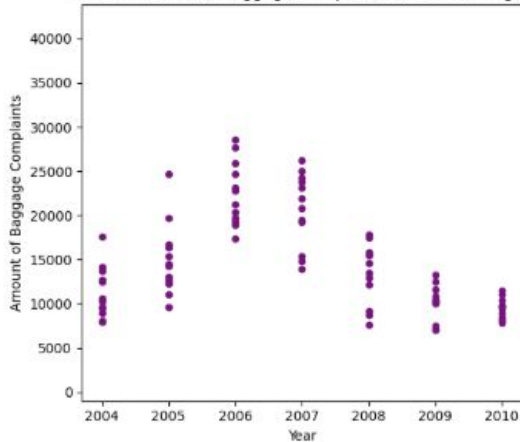
Although United has the most baggage complaints, American Eagle actually has more baggage complaints per number of passengers

Hawaiian

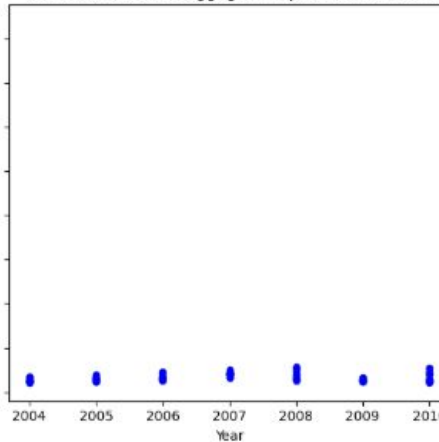




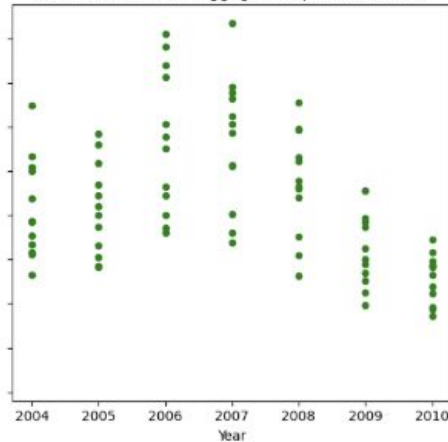
Year vs Amount of Baggage Complaints (American Eagle)



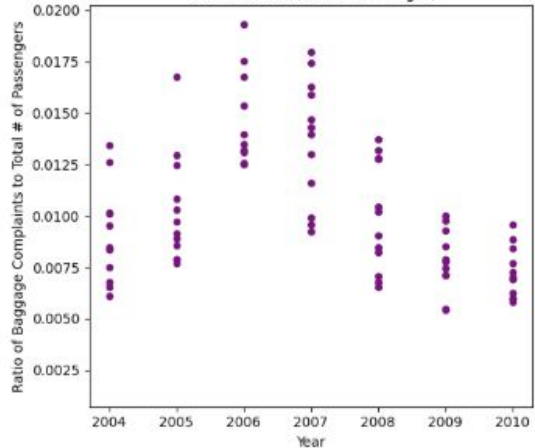
Year vs Amount of Baggage Complaints (Hawaiian)



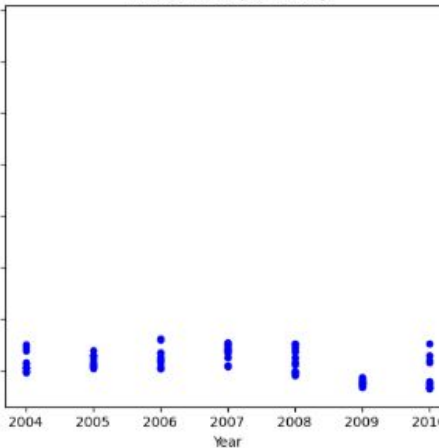
Year vs Amount of Baggage Complaints (United)



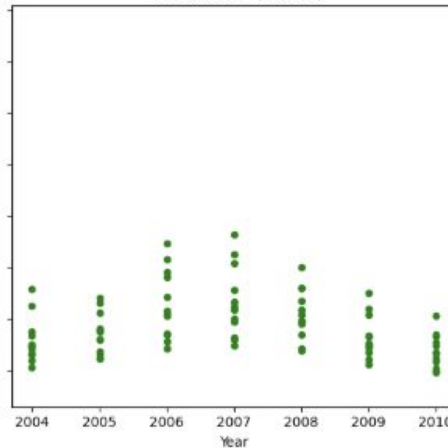
Year vs Ratio (American Eagle)



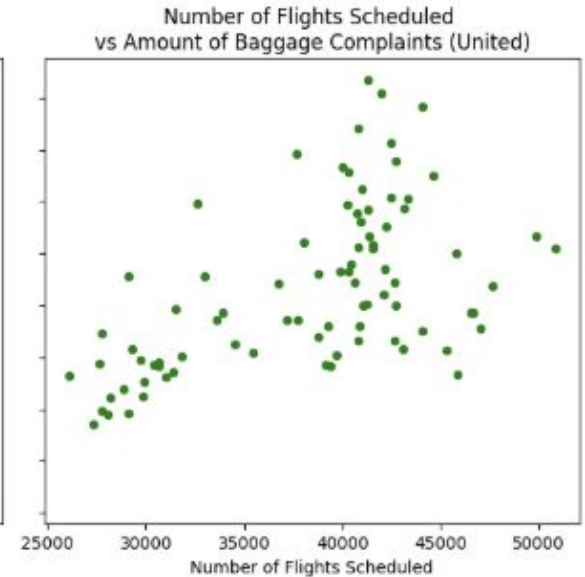
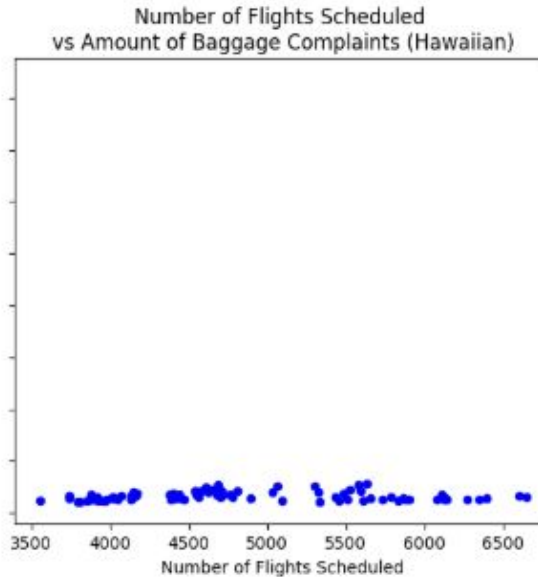
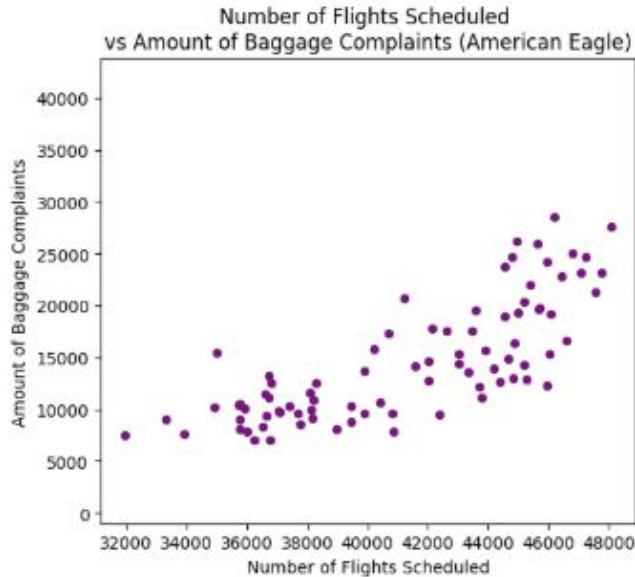
Year vs Ratio (Hawaiian)



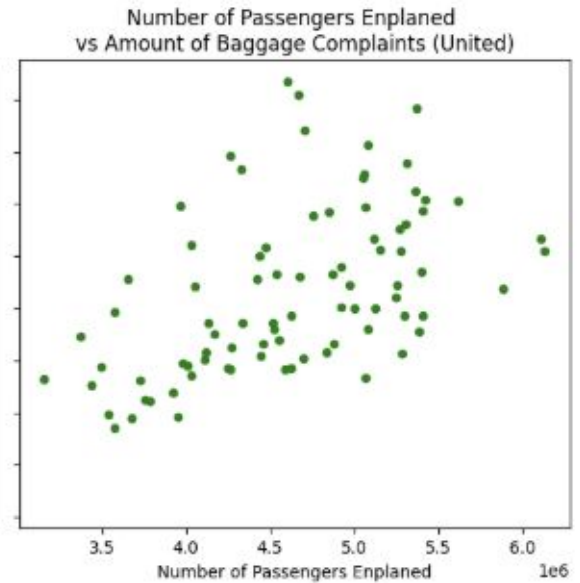
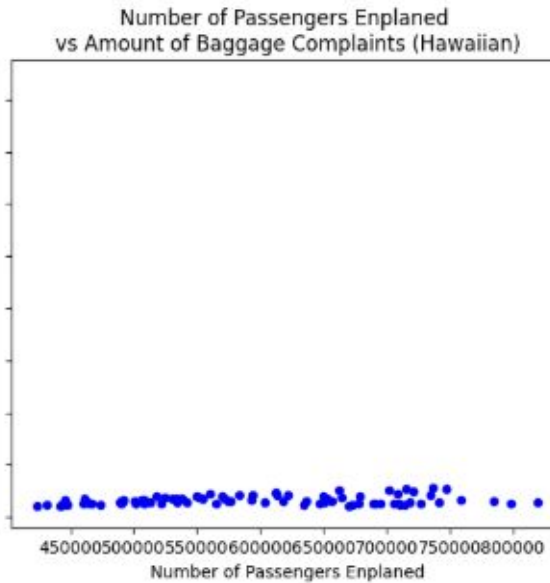
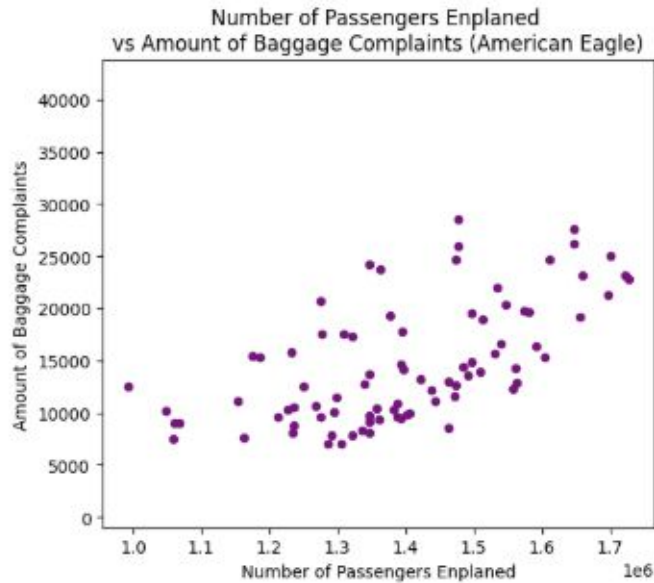
Year vs Ratio (United)



of Flights Scheduled vs # of Baggage Complaints

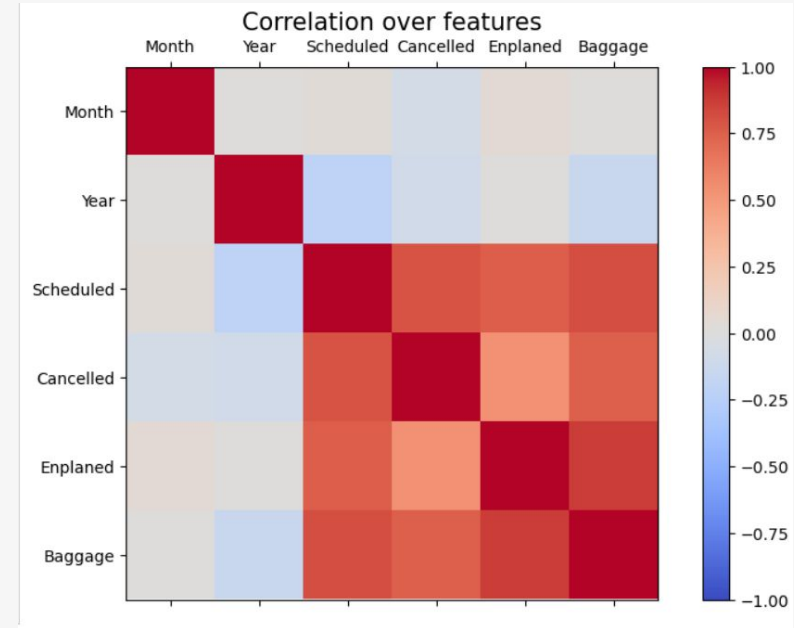


of Passengers Enplaned vs # of Baggage Complaints



Feature Correlations with # of Baggage Complaints

- Least correlated:
 - Month
 - Year
 - Most correlated:
 - Scheduled
 - Cancelled
 - Enplaned
- * this makes sense intuitively



02.5

Modeling Standards



Split Method: Chronological train/ test split

- **Why?**
 - Time-series data (future values should not influence model training)
 - Ensure model performance reflects forecasting scenario
 - train on historical data & predict the future
- **Why not use a random split?**
 - Random splitting mixes earlier and later months
 - Unreliable real-world predictions
- **How the Split was done**
 - Dataset was sorted by date from earliest to latest
 - First 75% of months used for training
 - Last 25% of months used for test
- **Benefits of a Chronological Split**
 - Prevents data leakage (future information leaking into training)
 - Makes the evaluation reflect real-world use

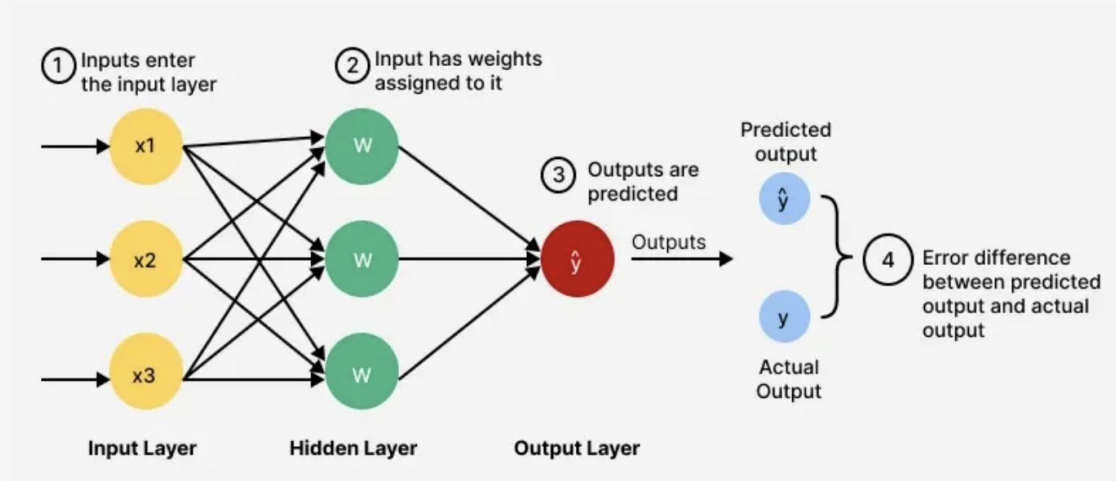
03

FFNN



FFNN - Feed Forward Neural Networks

- Neural Network where **data flows in a single direction** (i.e. from input to hidden layers to output)
- No loops or feedback
- Mainly used for pattern recognition tasks like image and speech classification



2 Approaches

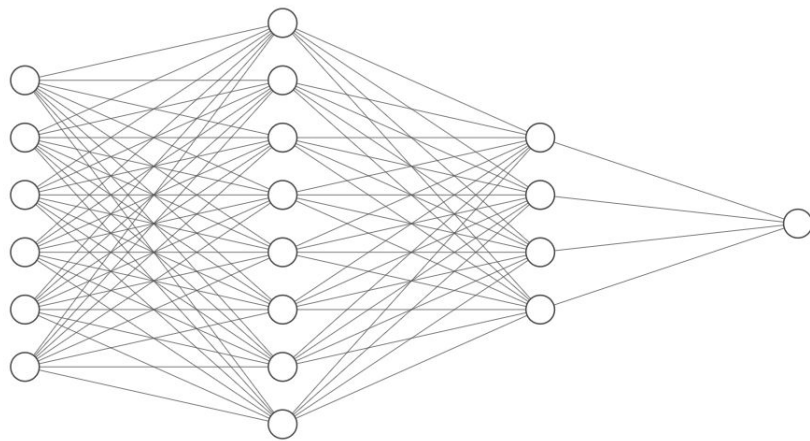
1. Not Separating by Airlines

- Includes airline as a feature
- Model sees data from all airlines at once and tries to learn patterns that generalize across them
- Airline identity was also included via one-hot encoding (AE, AH, AU)
- Benefit: More data for training
- Drawback: Less airline-specific precision

2. Separating by Airlines

- Filter dataset per airline and train a separate FFNN for each one
- Model sees only its airline's data
- Benefit: Airline-specific accuracy/ Model better for unique trends
- Drawback: Less data per model

1st Approach: Not separating by airlines



Input Layer $\in \mathbb{R}^6$

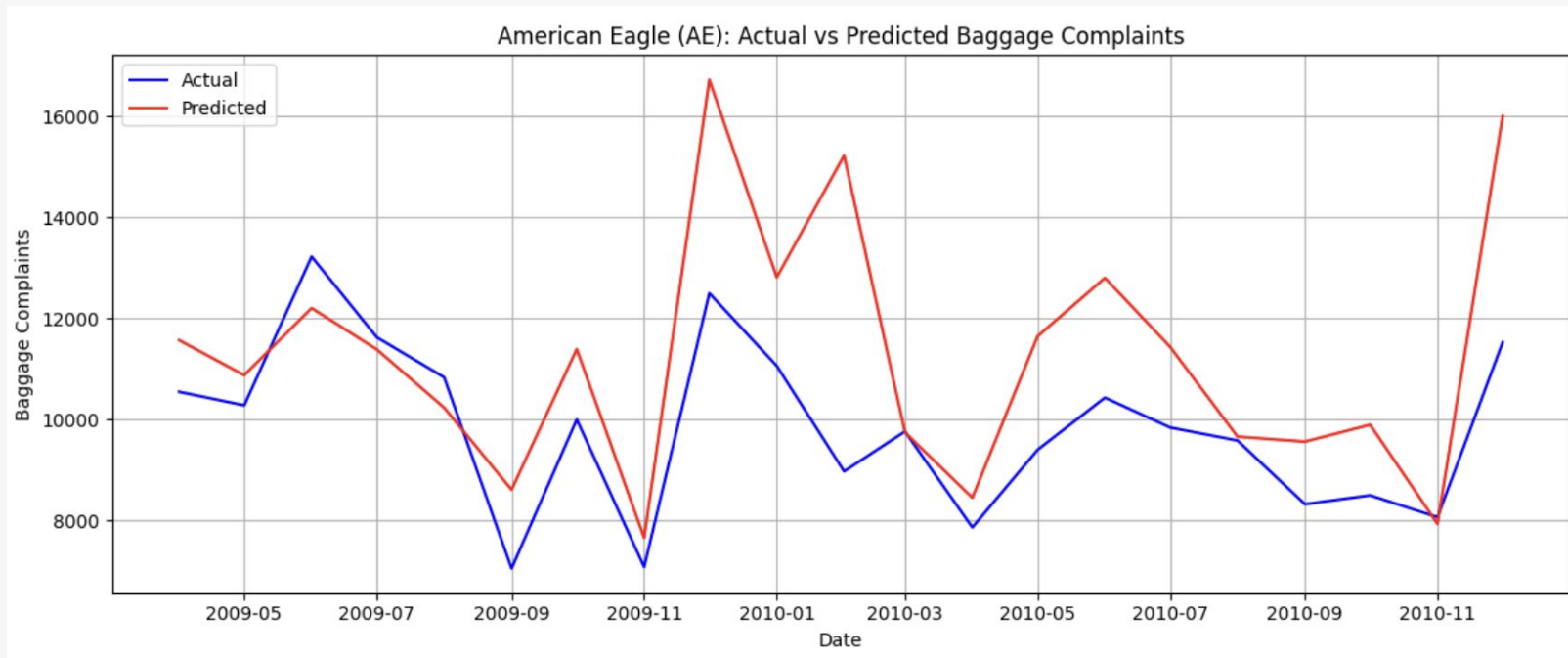
Hidden Layer $\in \mathbb{R}^8$

Hidden Layer $\in \mathbb{R}^4$

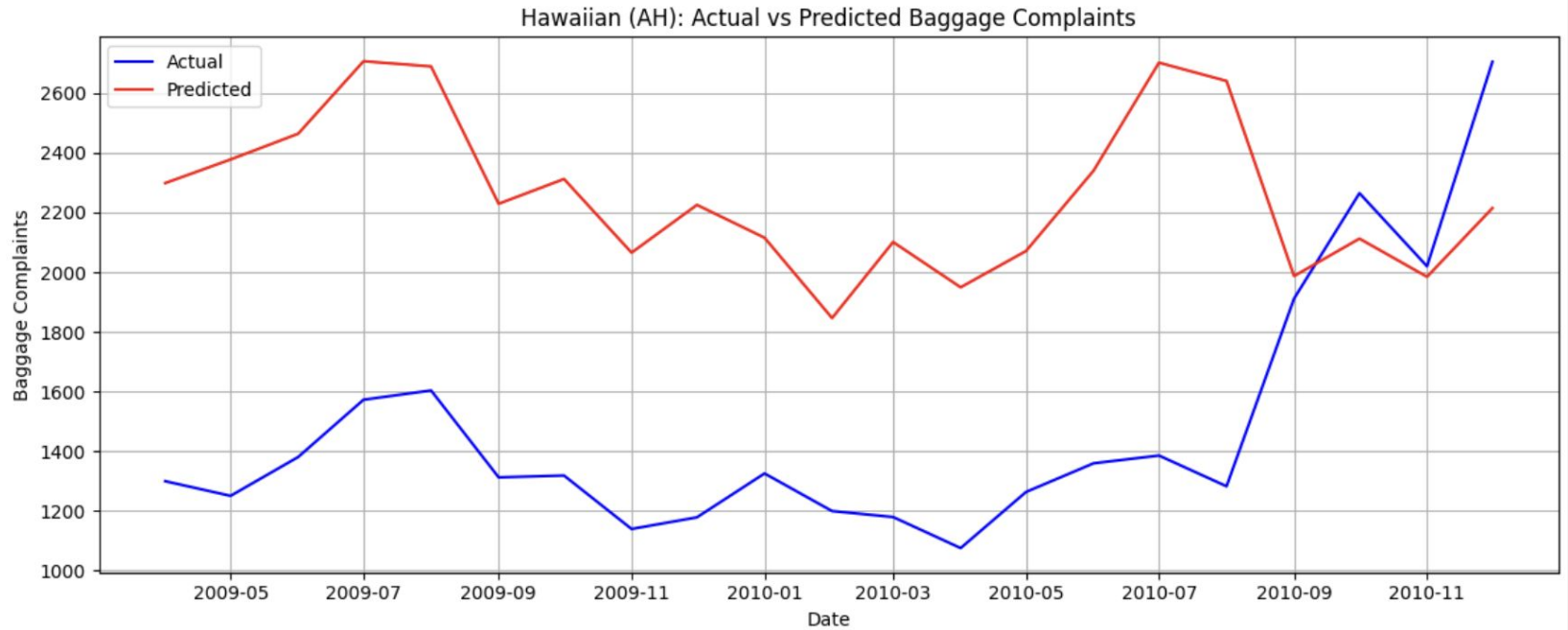
Output Layer $\in \mathbb{R}^1$

- **6 inputs** (Scheduled, Cancelled, Enplaned, AE, AH, AU)
- **2 Hidden layers**
 - First hidden layer: 8 neurons
 - Second hidden layer: 4 neurons
- **1 output** (# of Baggage Complaints)

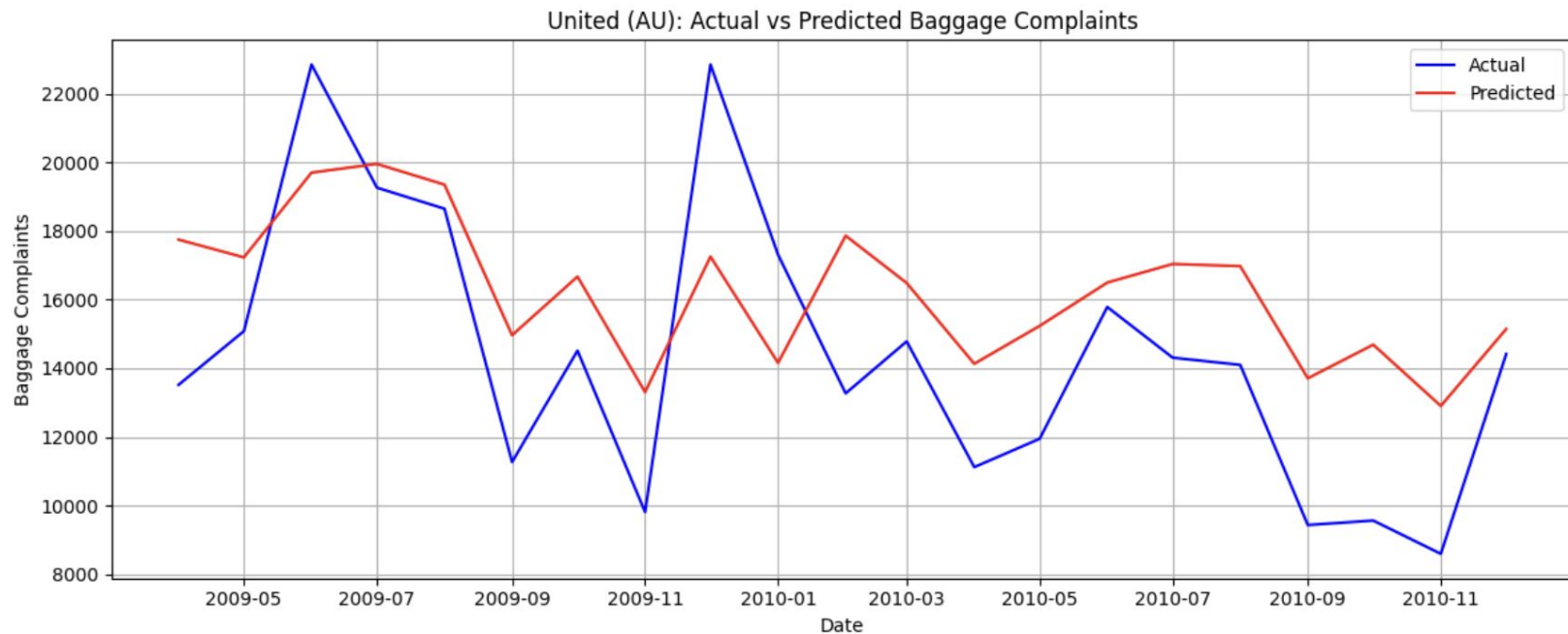
Not Separating by Airlines: American Eagle



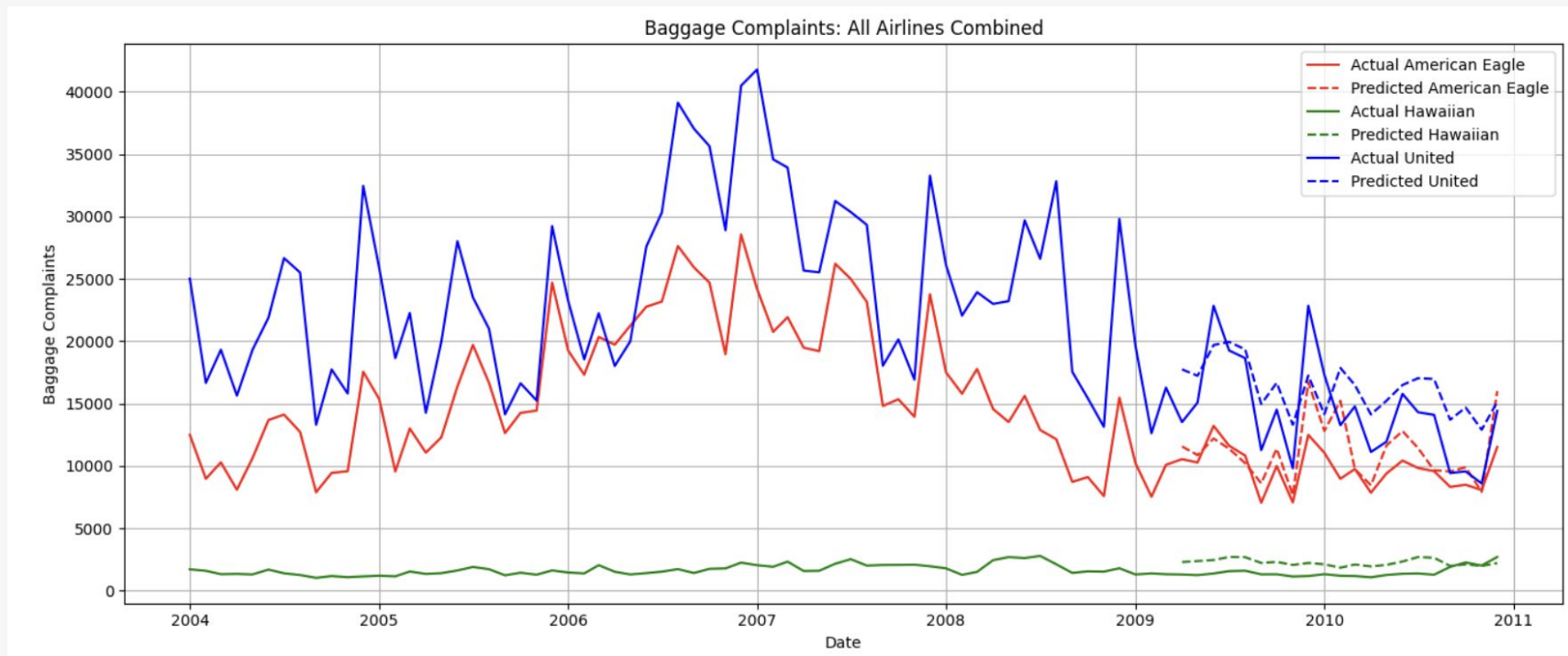
Not Separating by Airlines: Hawaiian



Not Separating by Airlines: United

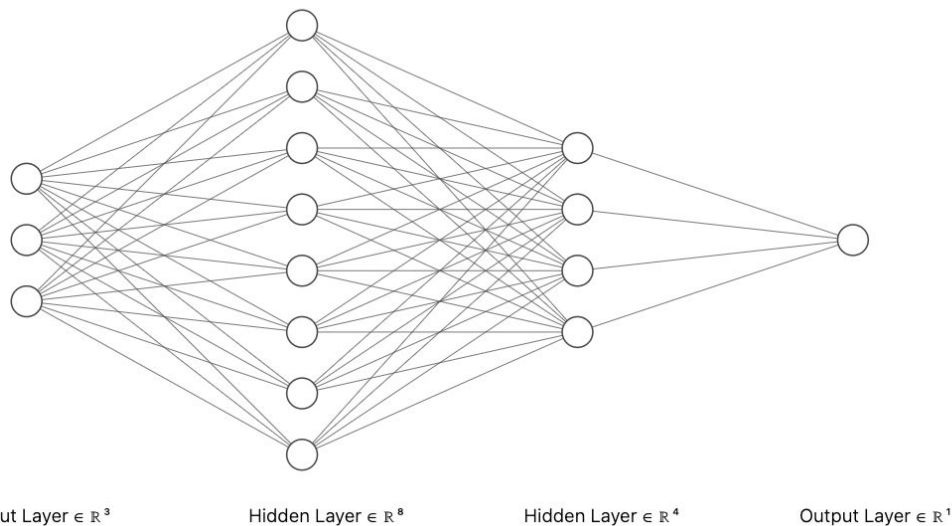


Not Separating by Airlines - Full Graph



Test RMSE: 2360.41

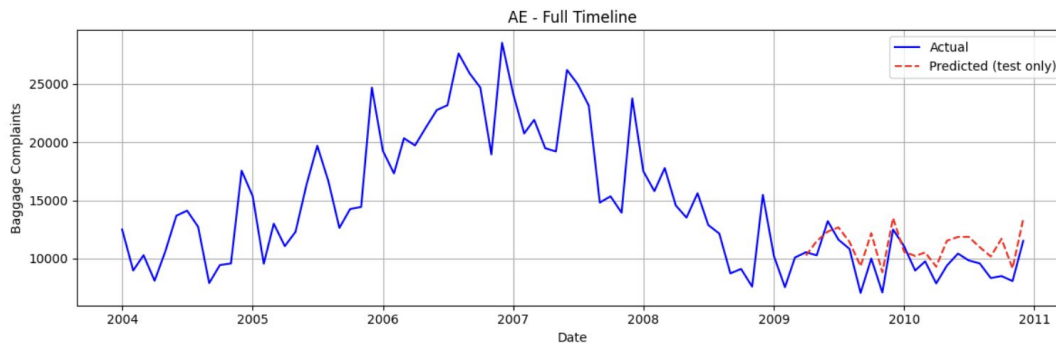
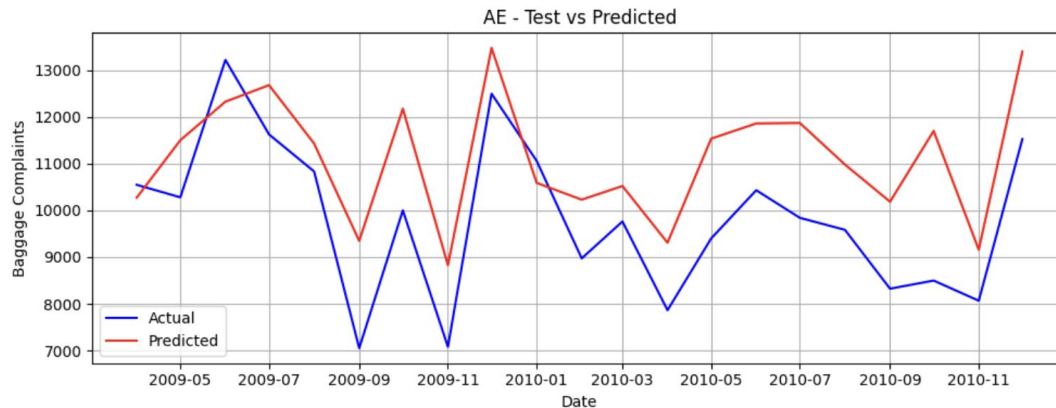
2nd Approach: Separating by airlines



For each Airline:

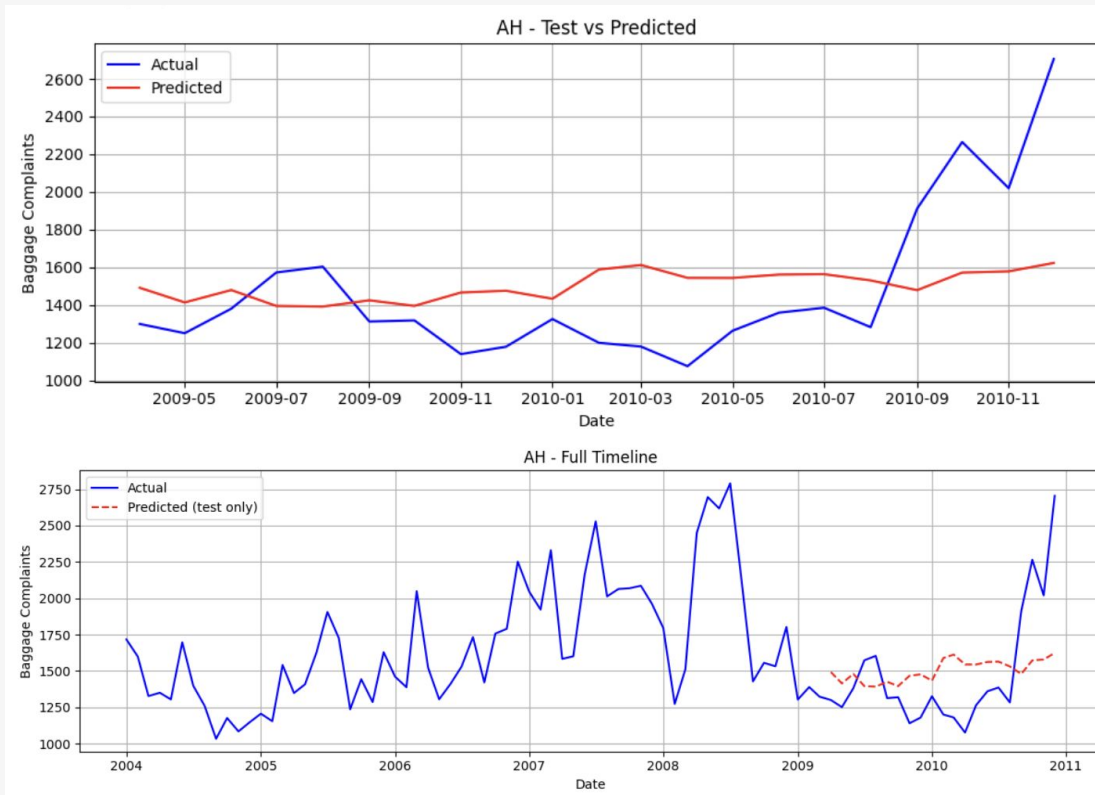
- **3 inputs** (Scheduled, Cancelled, Enplaned)
- **2 Hidden layers**
 - First hidden layer: 8 neurons
 - Second hidden layer: 4 neurons
- **1 output** (# of Baggage Complaints)

Separating by Airlines: American Eagle



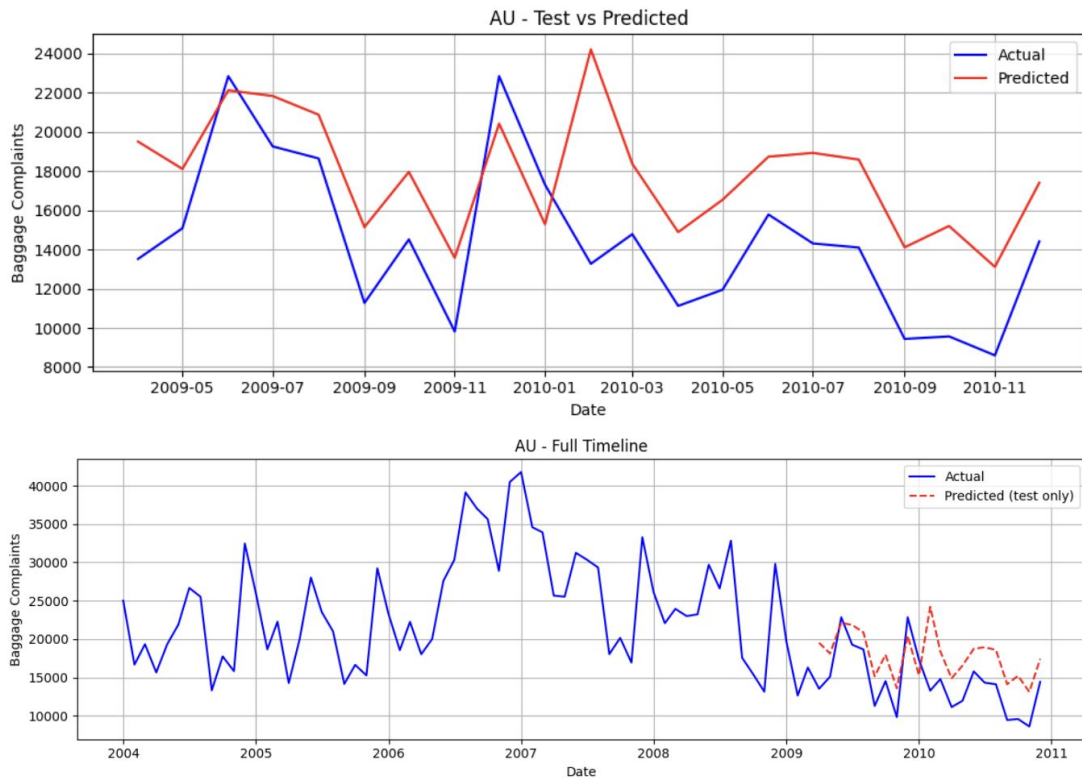
Test RMSE (AE): 1594.51

Separating by Airlines: Hawaiian



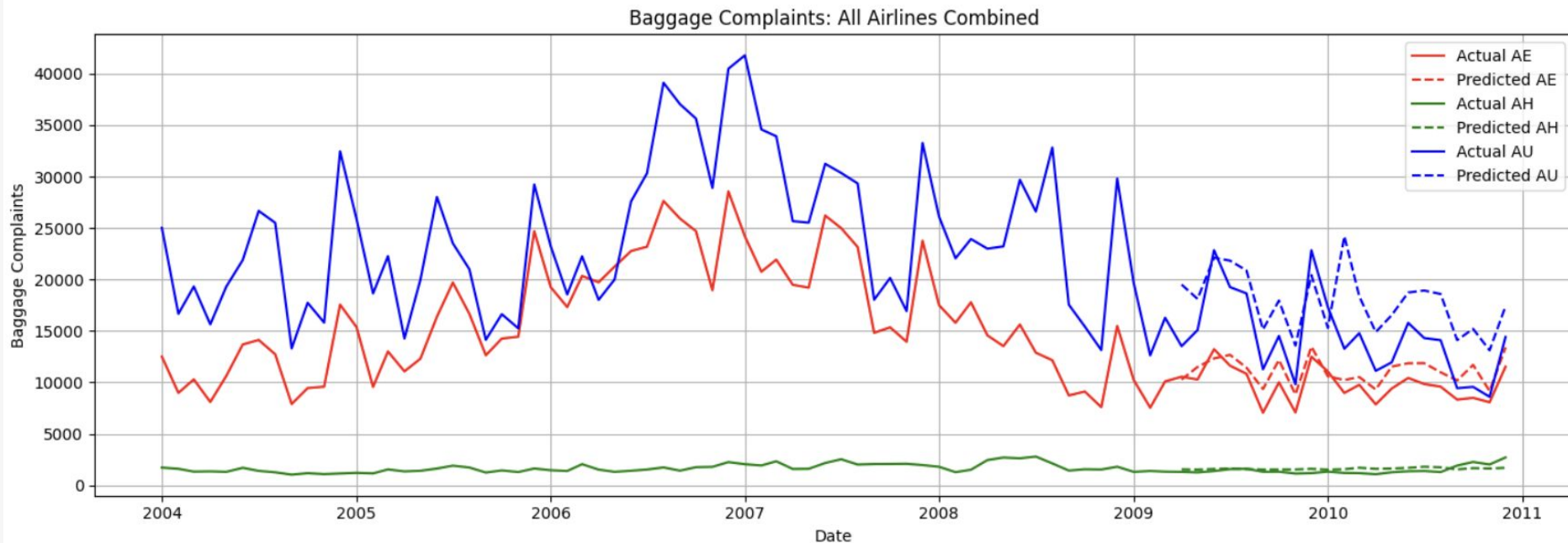
Test RMSE (AH): 388.89

Separating by Airlines: United



Test RMSE (AU): 4412.91

Separating by Airlines - Full Graph



Test RMSE (AE): 1594.51 Test RMSE (AH): 388.89 Test RMSE (AU): 4412.91

Average RMSE: 2132.103

FFNN Conclusions

- **1st Approach:**
 - RMSE: 2360.41
- **2nd Approach:**
 - Average: 2132.103
 - AE: 1594.51
 - U: 4412.91
 - H: 388.89



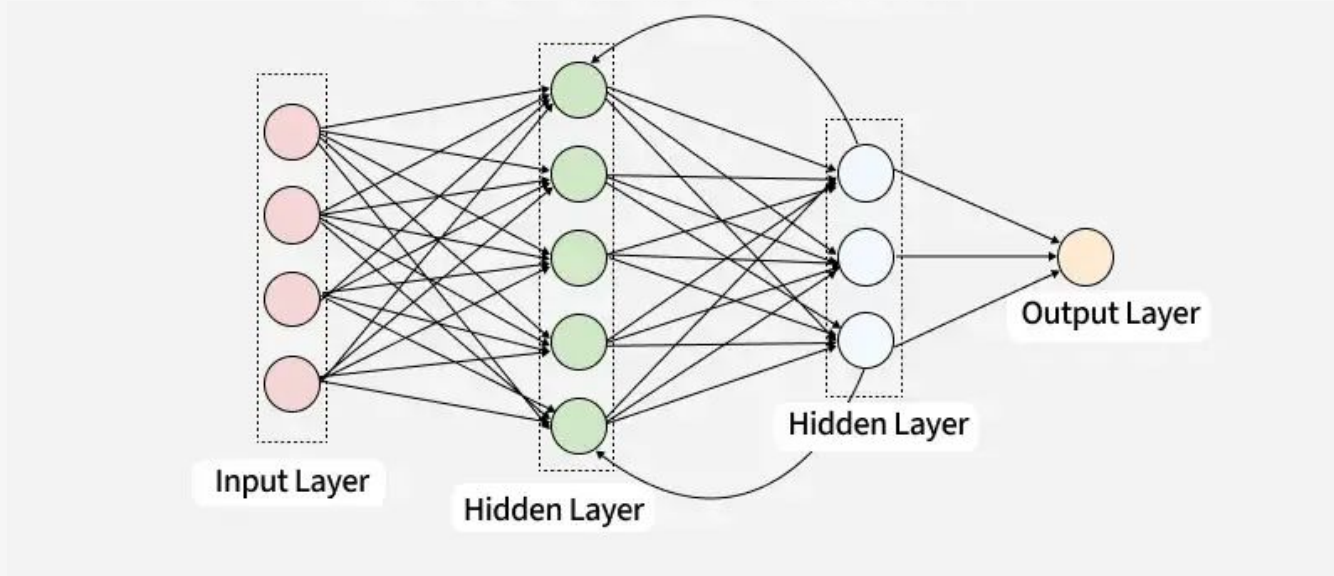
04

RNN

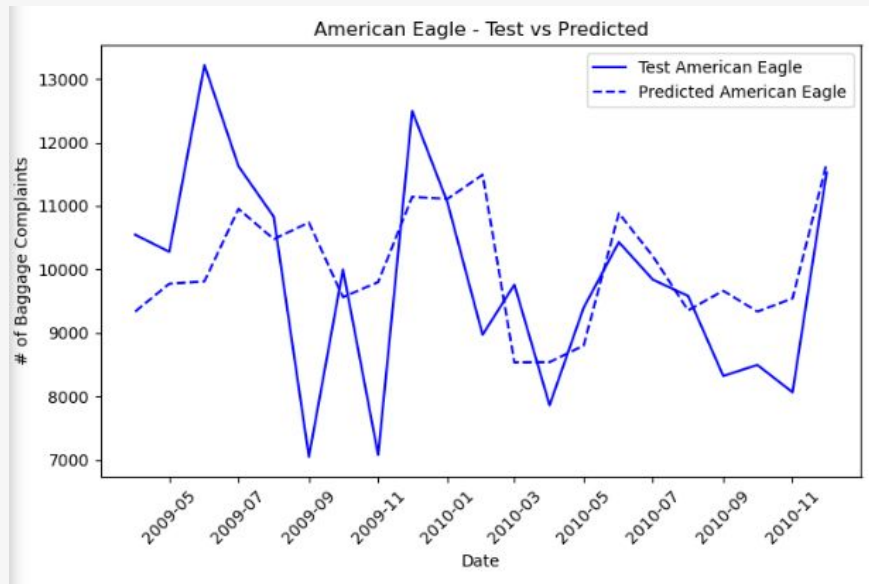
RNN - Recurrent Neural Network

- Used to process **sequential data** (data with temporal order)
 - Simple benchmark to compare against more complex models like LSTM
 - Few parameters
 - Good for short-term dependencies
 - Short-term memory problem
 - Gradient **explodes** (become too large) or **vanishes** (become too small)

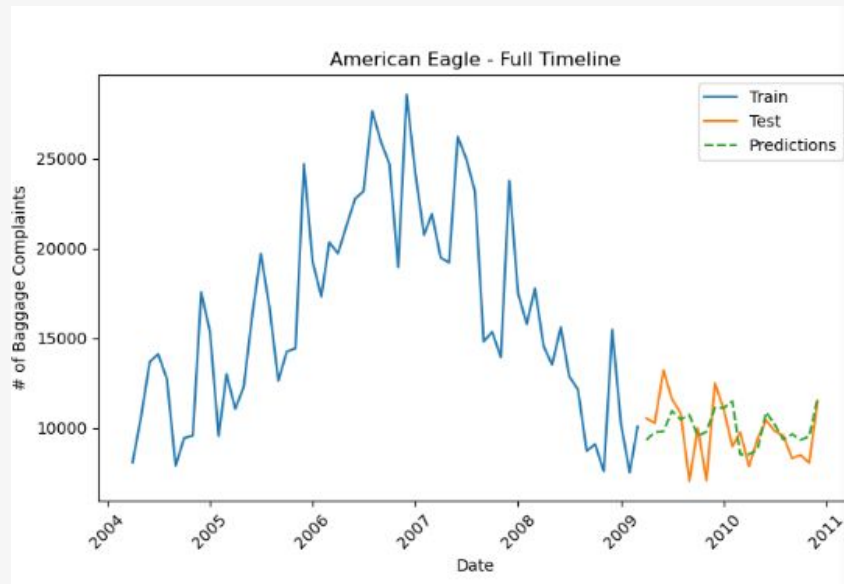
RNN - Recurrent Neural Network



Separating Airlines: American Eagle

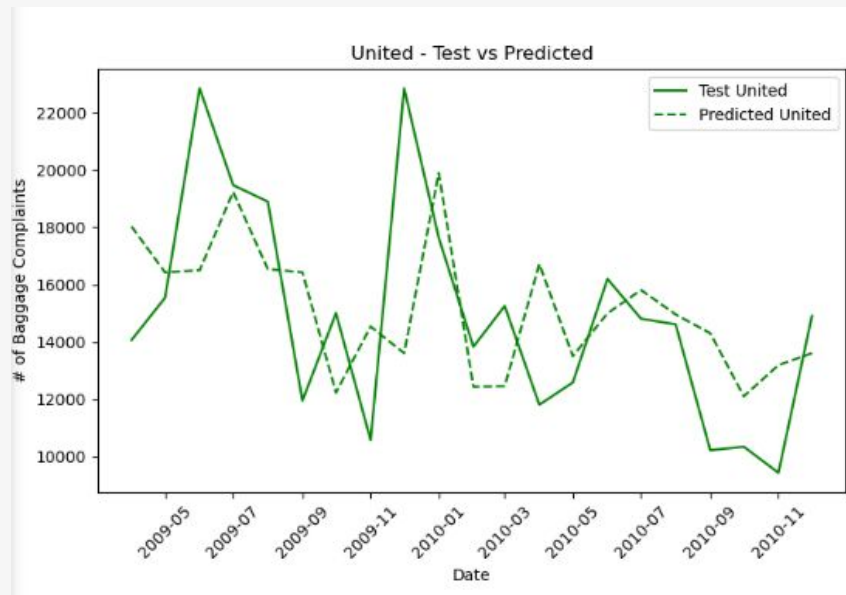


AE Test RMSE: 1368.829

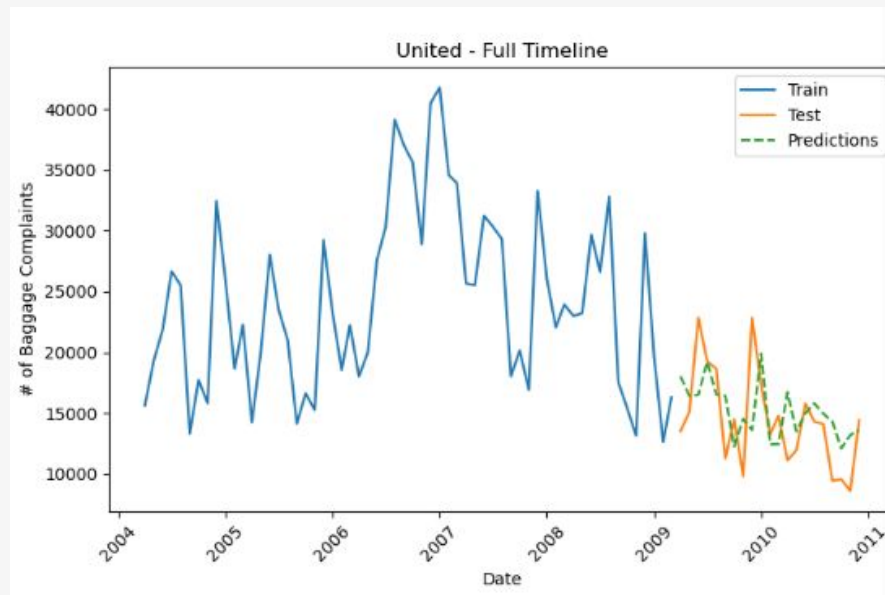


AE Train RMSE: 2216.613

Separating Airlines: United

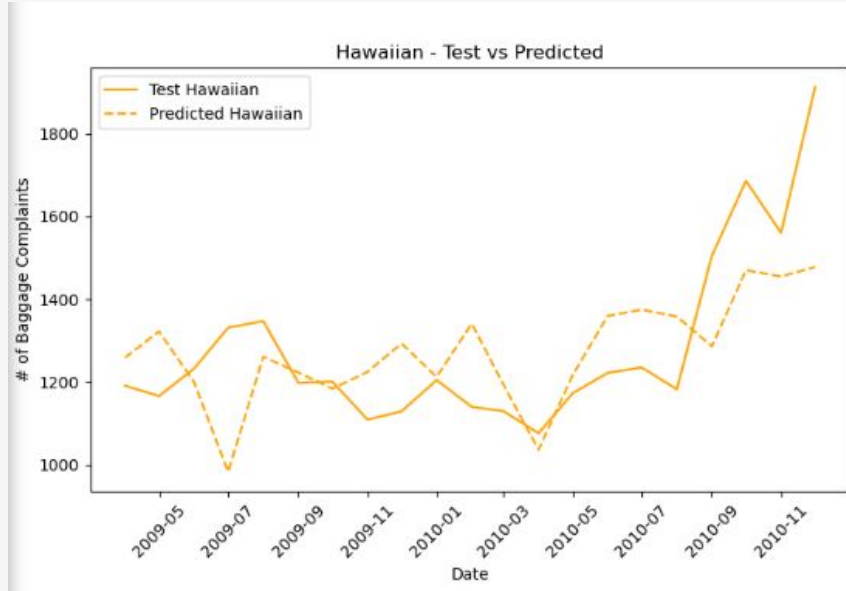


United Test RMSE: 2991.098

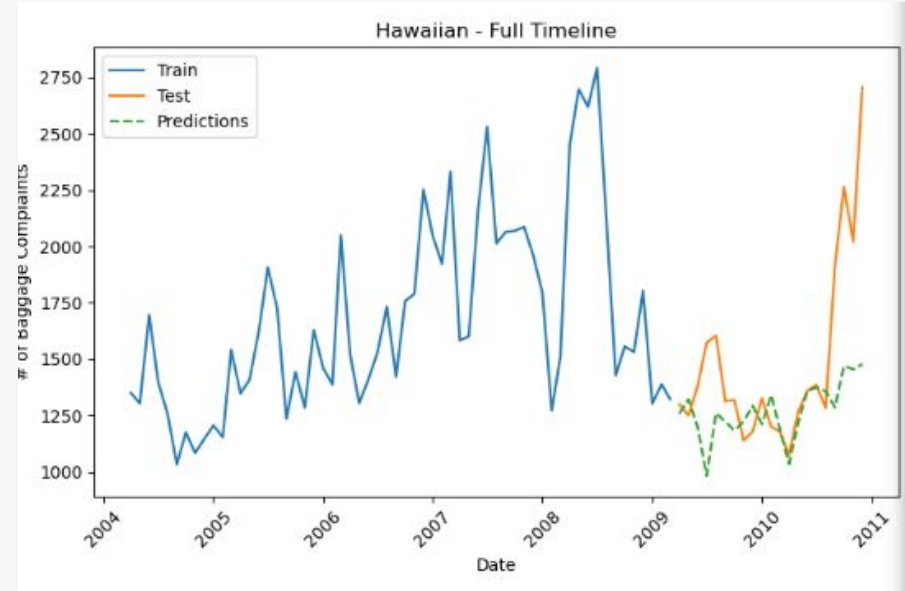


United Train RMSE: 4615.048

Separating Airlines: Hawaiian

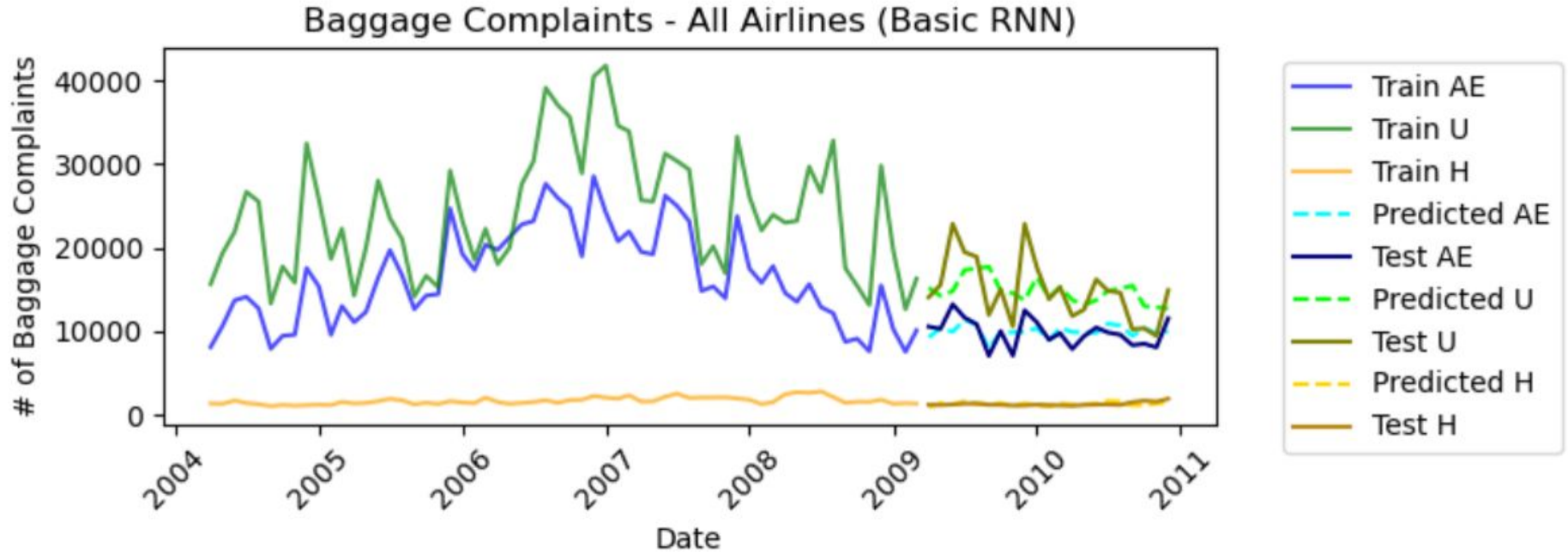


Hawaiian Test RMSE: 150.353



Hawaiian Train RMSE: 310.435

Separating Airlines: Combined Plot



Test RMSE: 1368.829 Test RMSE: 2991.098 Test RMSE: 150.353 Average RMSE: 1503.427

RNN Conclusions

- Faster training and lower memory usage
- Quick to forget past information
- Seasonality trends such as holiday peaks and summer vacation peaks in a basic RNN are not as clear
- Might also miss longer-term patterns
- Too basic for Approach 1 (not splitting by airline) → patterns would become simple and unique airline-specific trends may be lost

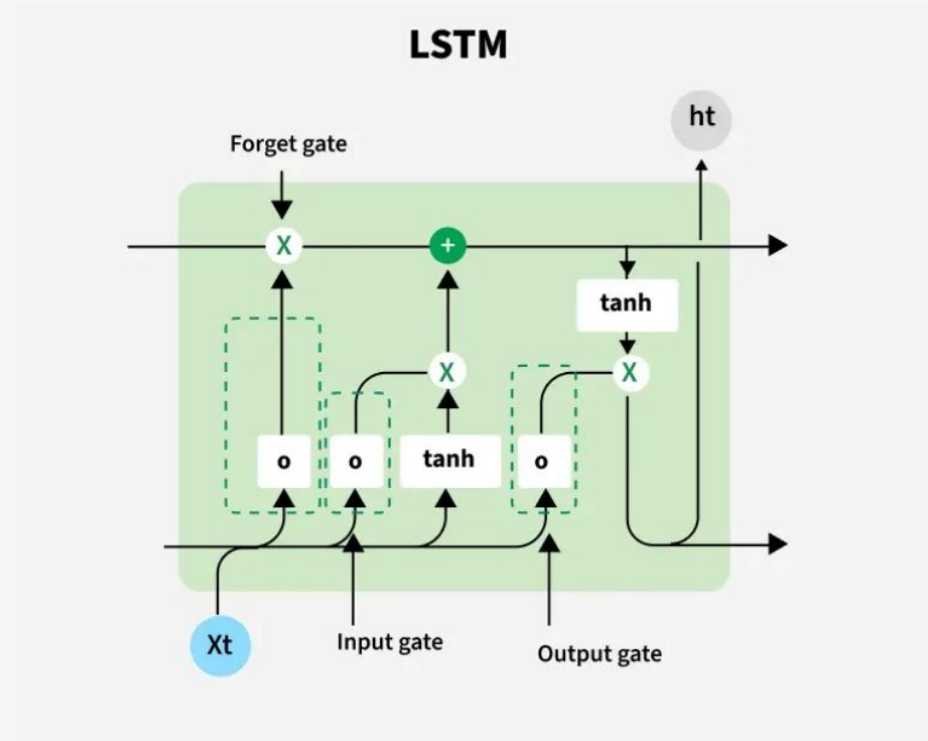
05

LSTM



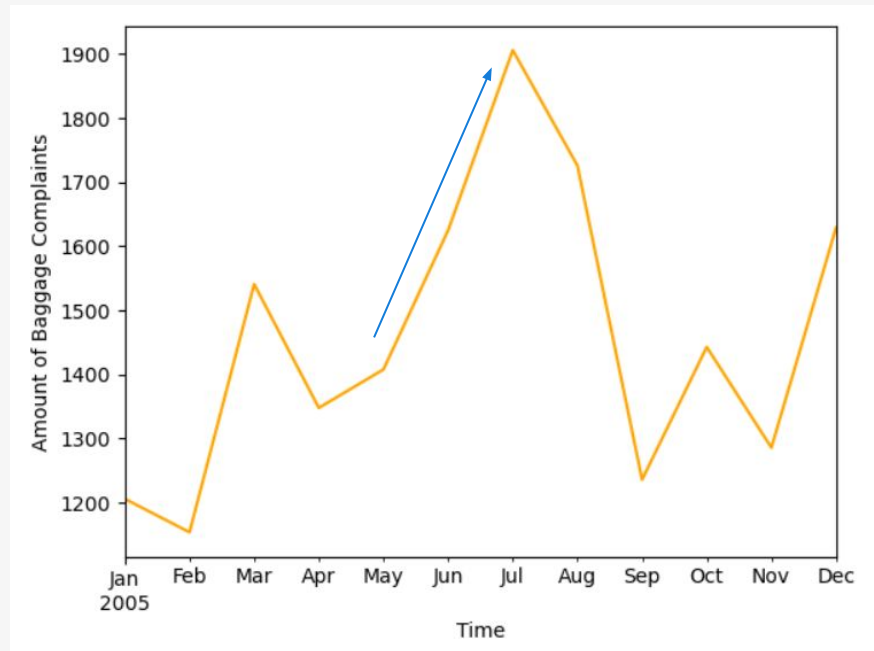
LSTM - Long Short-Term Memory

- An improved form of RNN
 - Solves the “vanishing gradient problem”
 - Input Gate
 - Forget Gate
 - Output Gate
 - Slower to train and more demanding in memory



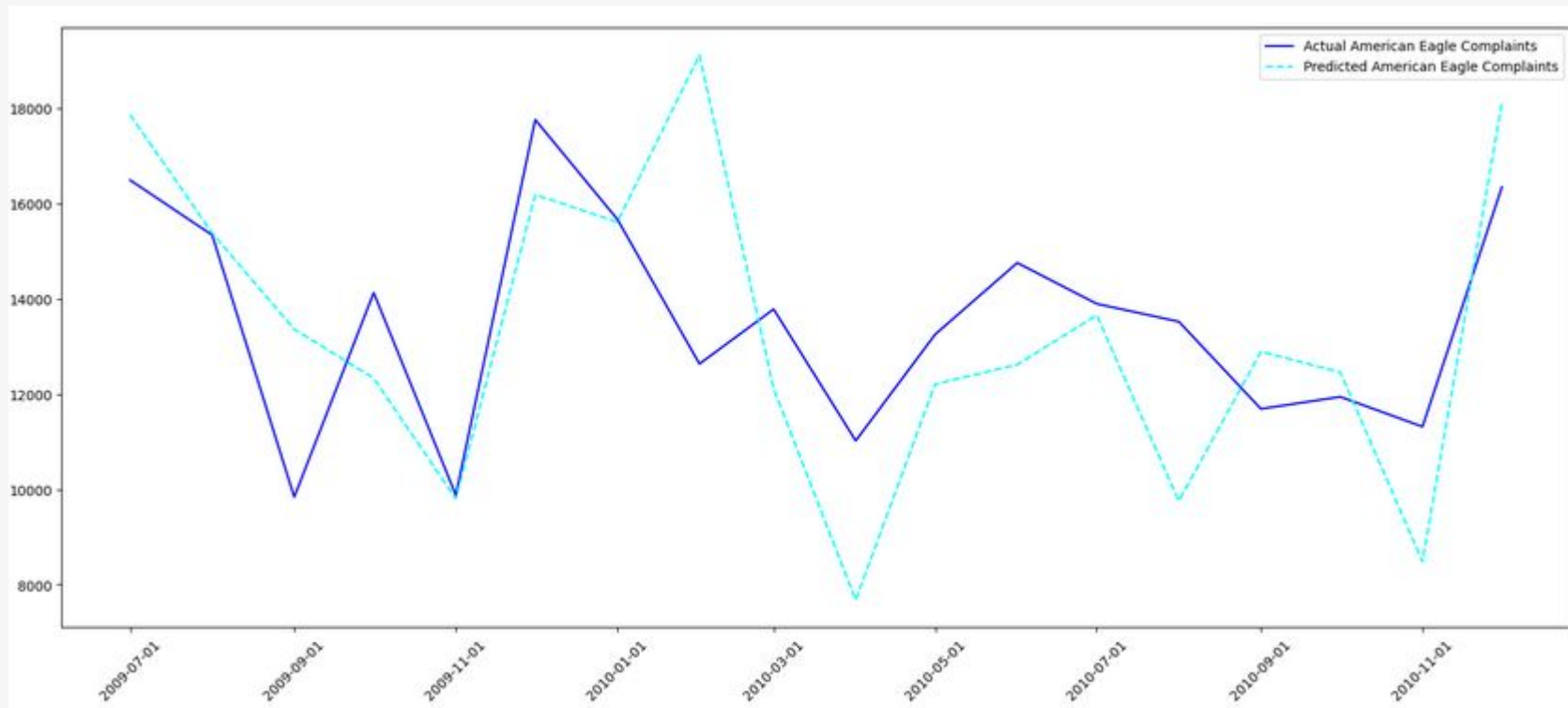
Approach 1: Not Separating Airlines

	Baggage	Scheduled	Cancelled	Enplaned	AE	AH	AU
Date							
2004-01-01	12502	38276	2481	992360	True	False	False
2004-02-01	8977	35762	886	1060618	True	False	False
2004-03-01	10289	39445	1346	1227469	True	False	False
2004-04-01	8095	38982	755	1234451	True	False	False
2004-05-01	10618	40422	2206	1267581	True	False	False
2004-06-01	13684	39879	1580	1347303	True	False	False
2004-07-01	14121	41586	1819	1396642	True	False	False
2004-08-01	12732	42016	1465	1339264	True	False	False
2004-09-01	7895	40871	1077	1292147	True	False	False
2004-10-01	9444	42381	1090	1393881	True	False	False
2004-11-01	9582	40839	1568	1274530	True	False	False
2004-12-01	17555	42629	1514	1308685	True	False	False

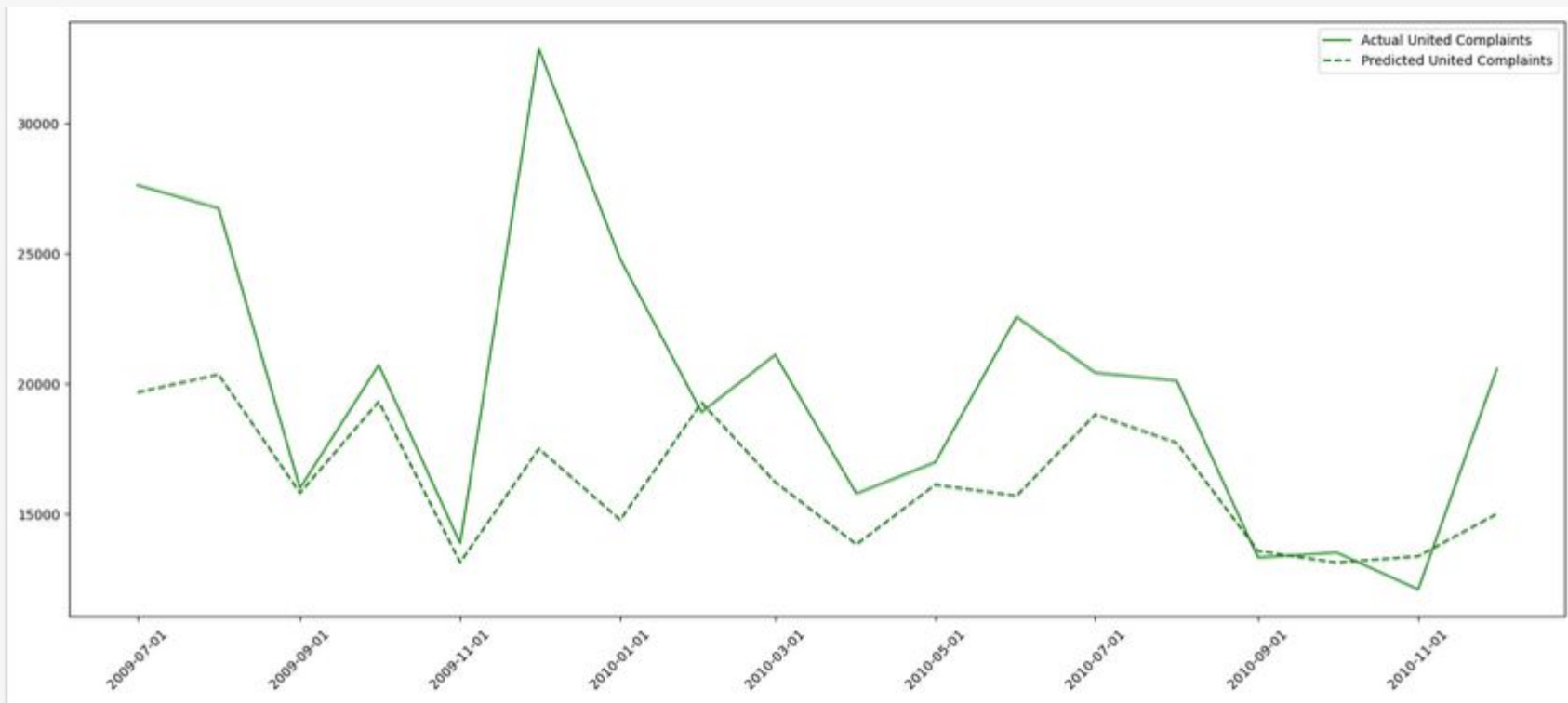


Seasonal!

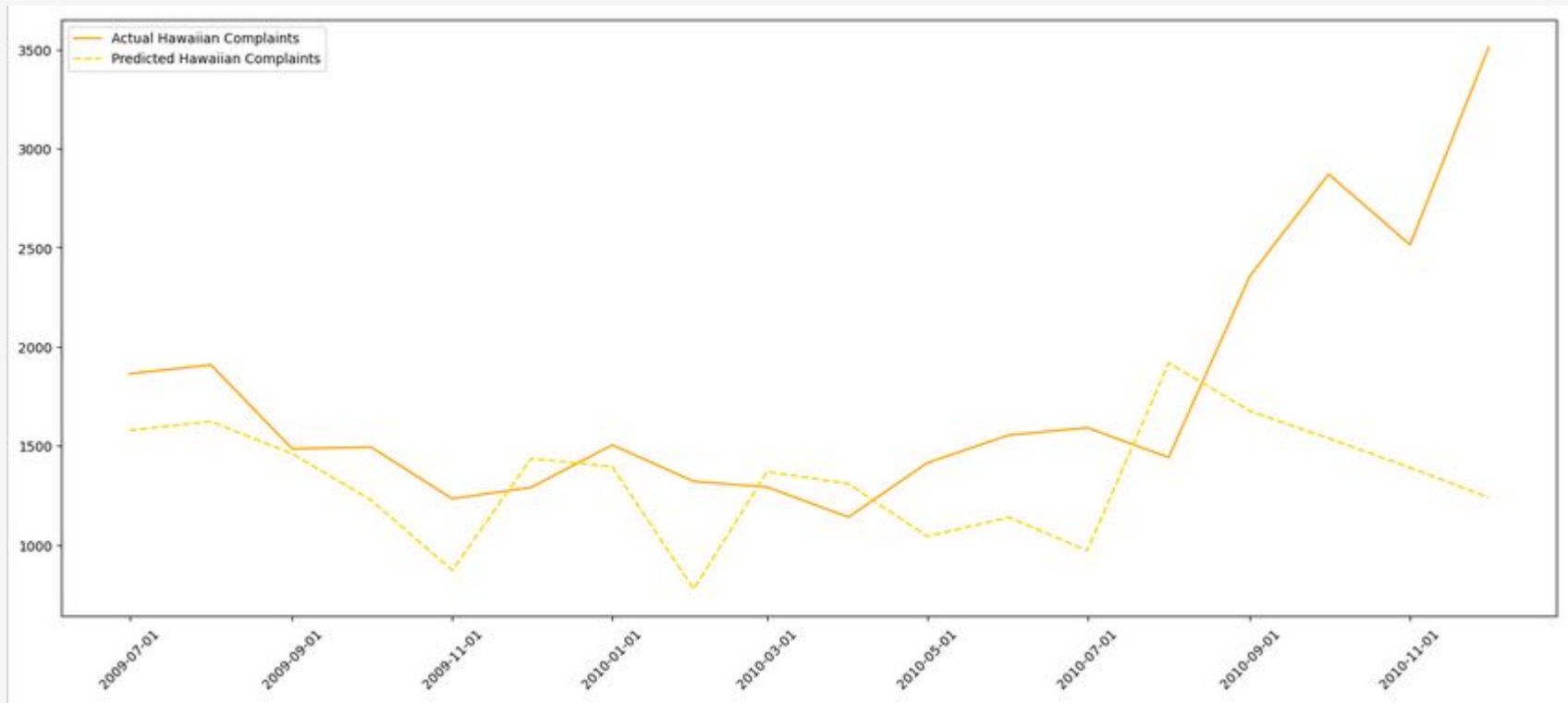
Approach 1: Not Separating Airlines



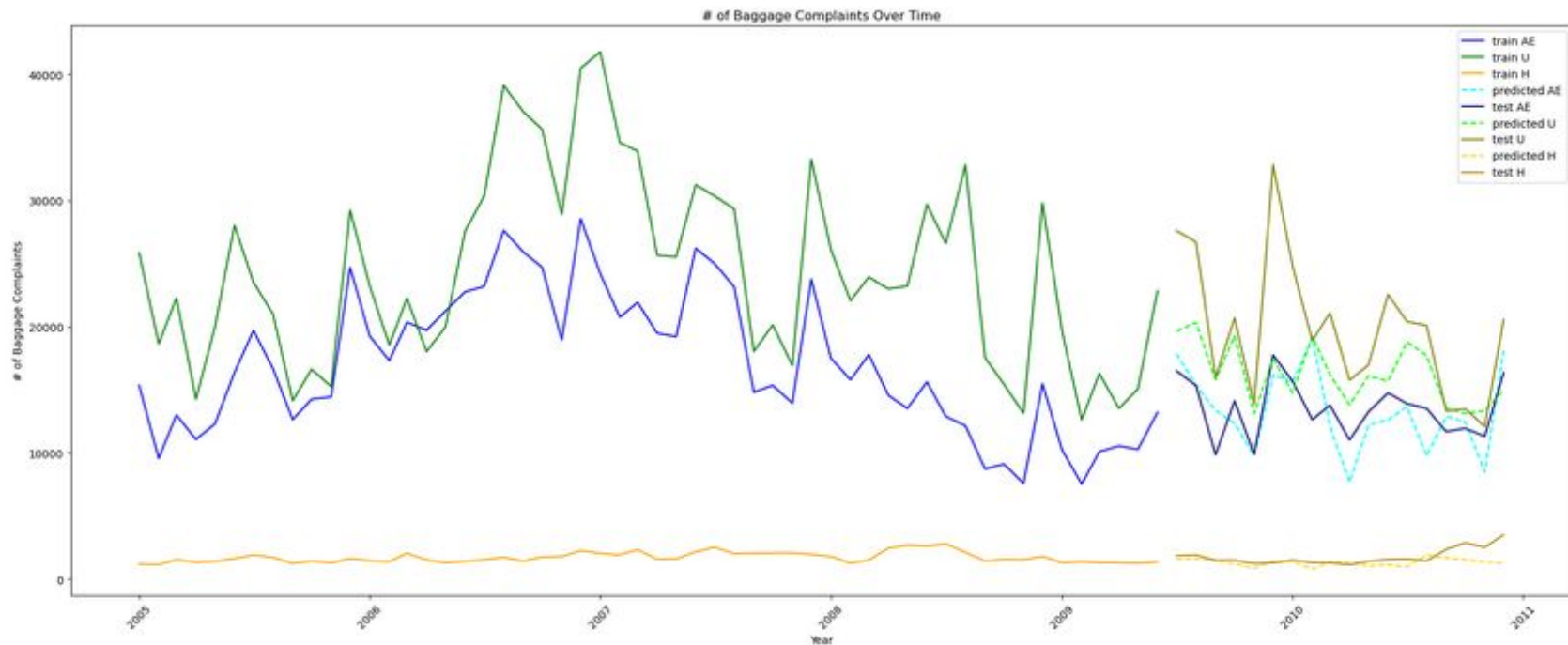
Approach 1: Not Separating Airlines



Approach 1: Not Separating Airlines



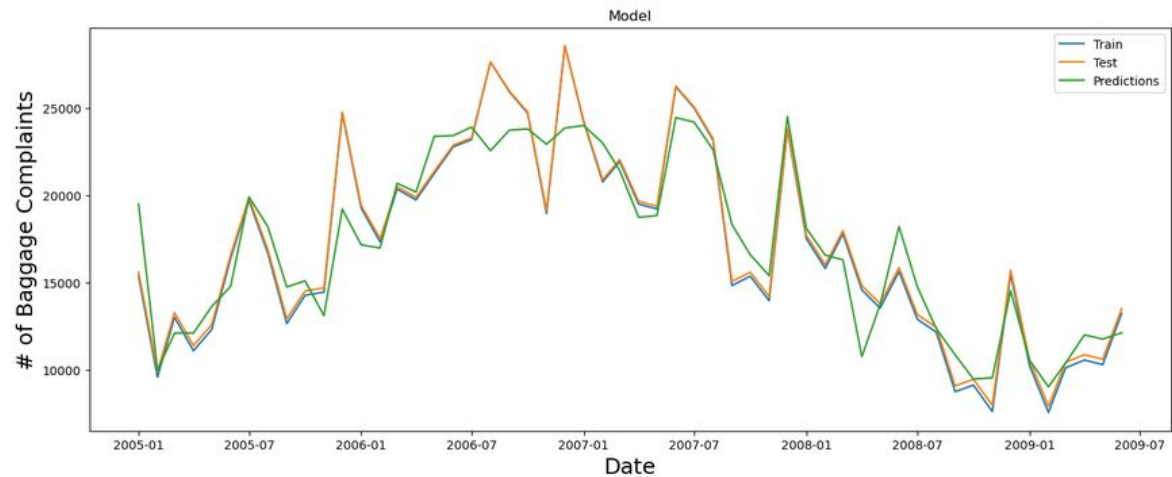
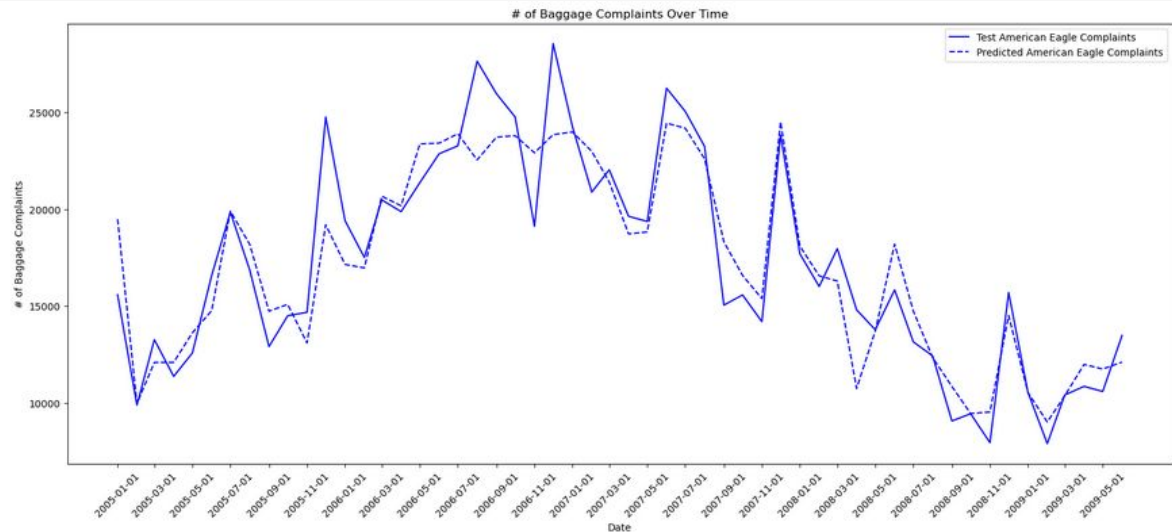
Approach 1: Not Separating Airlines



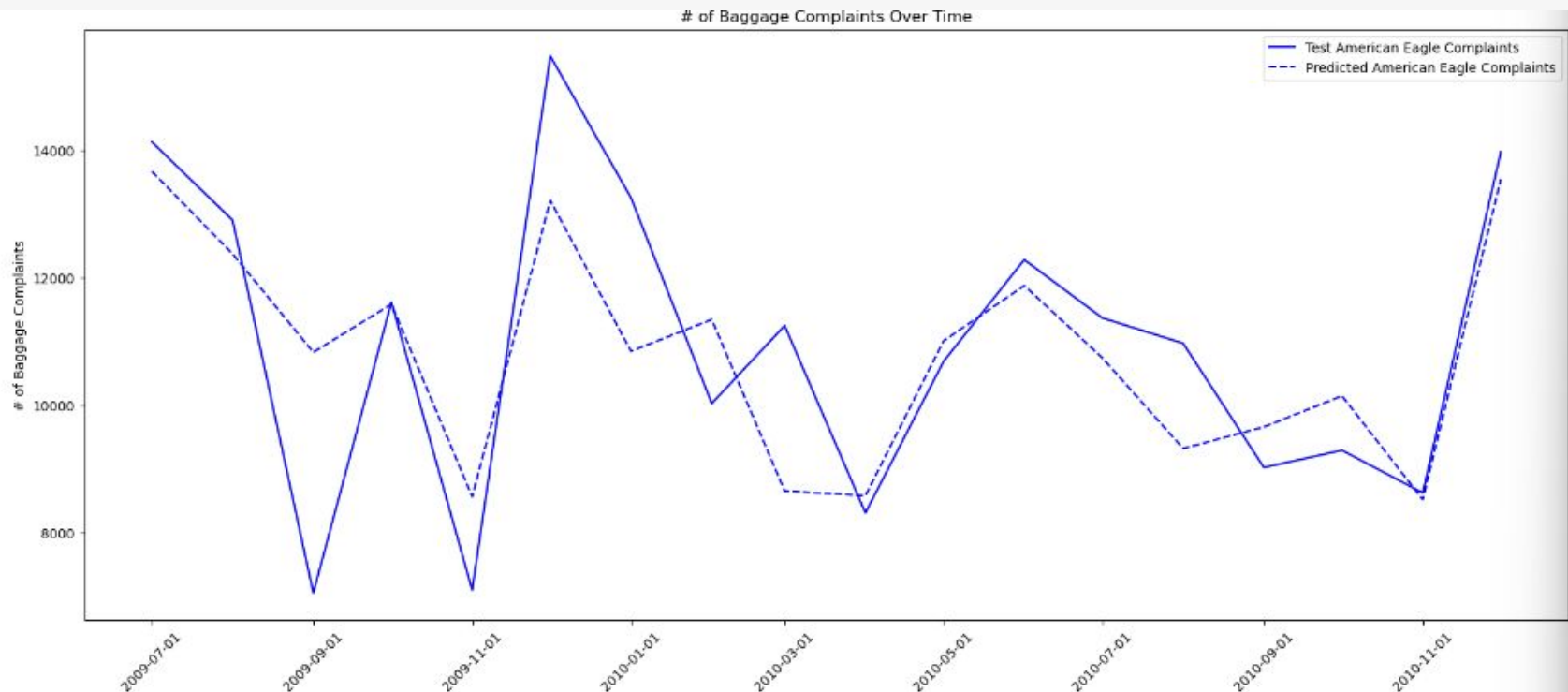
Test RMSE: 3544.094

Approach 2: Separating Airlines

Train RMSE: 1943.164

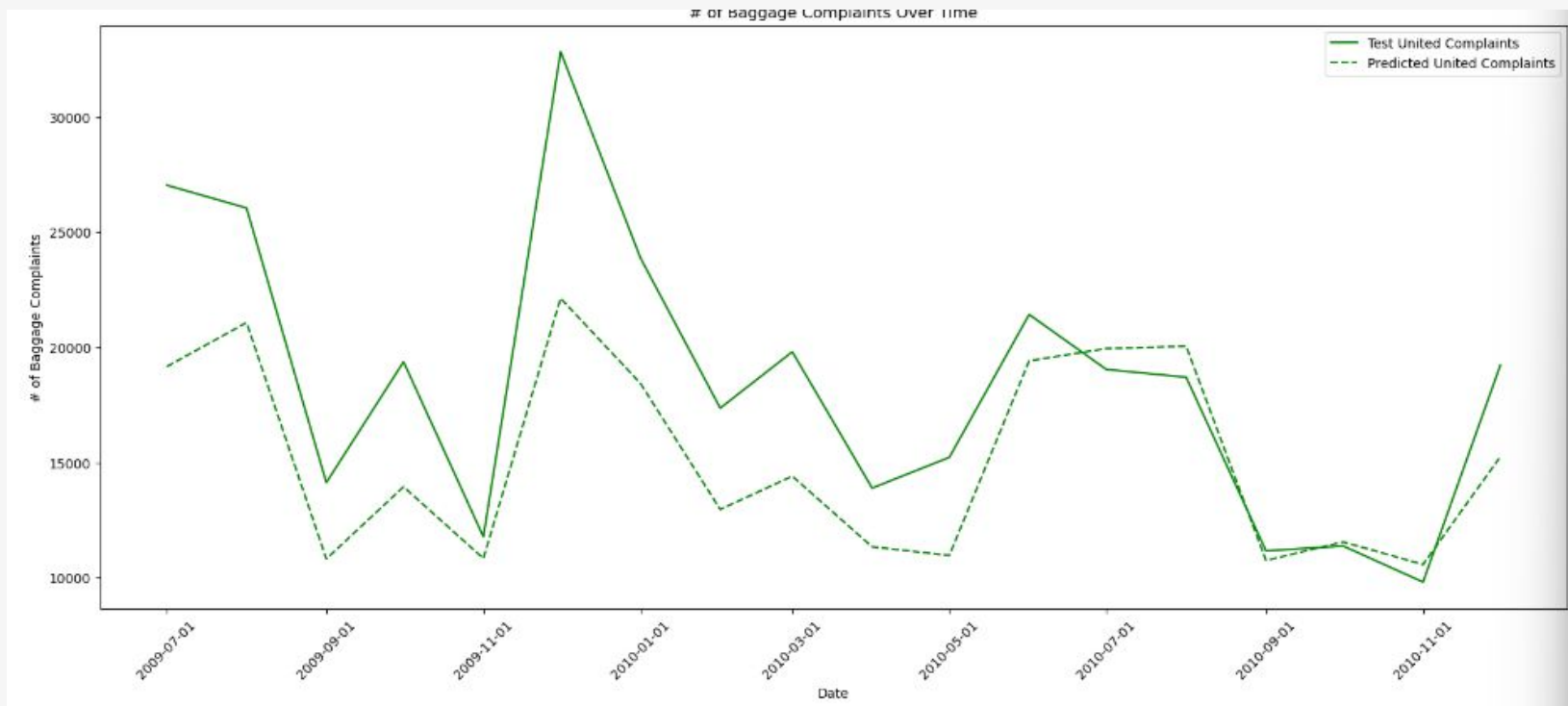


Approach 2: Separating Airlines



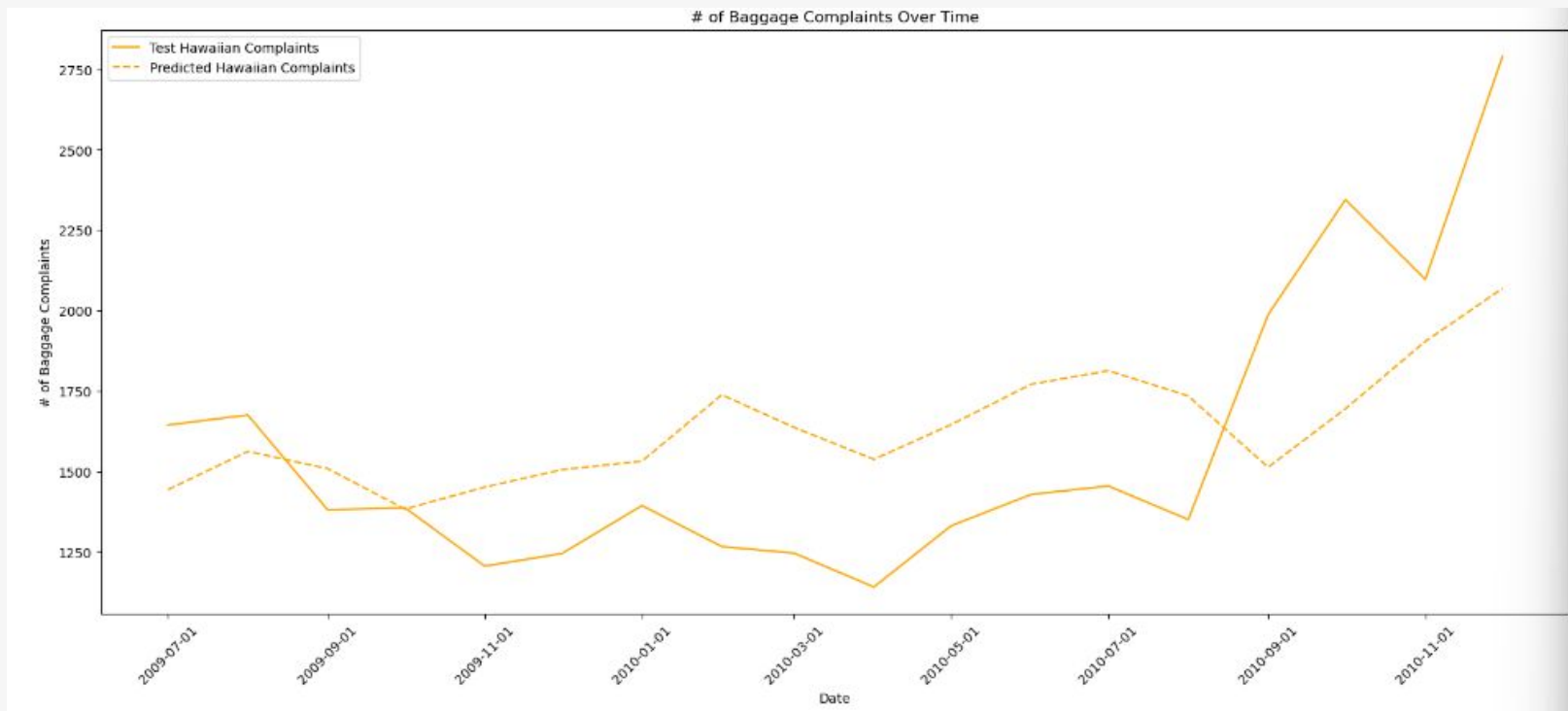
Test RMSE: 1510.557

Approach 2: Separating Airlines



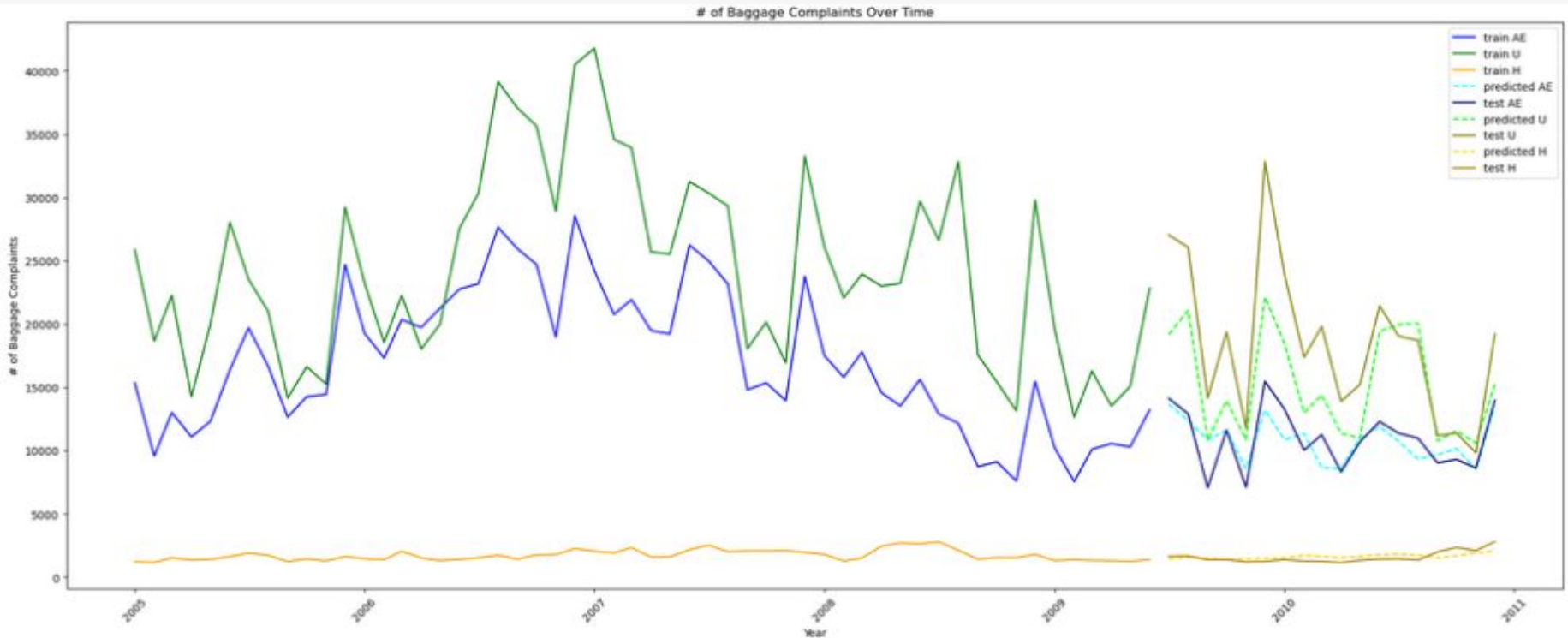
Test RMSE: 4523.791

Approach 2: Separating Airlines



Test RMSE: 368.996

Approach 2: Separating Airlines



Test RMSE: 1510.557 Test RMSE: 4523.791 Test RMSE: 368.996 Average RMSE: 2134.448

06

Conclusion



Results: RMSE

Median Value

	Month	Year	Baggage	Scheduled	Cancelled	Enplaned
Airline						
American Eagle	6.5	2007.0	13111.0	42021.0	1341.0	1391112.5
Hawaiian	6.5	2007.0	1516.5	4681.0	11.0	580140.0
United	6.5	2007.0	19986.5	40330.5	617.5	4628089.0

FFNN

- First Approach
2360.41
- Second Approach
 - Average:
2132.103
 - AE: 1594.51
 - U: 4412.91
 - H: 388.89

RNN

- Second Approach
 - Average:
1503.427
 - AE: 1368.829
 - U: 2991.098
 - H: 150.353

LSTM

- First Approach
 - 3544.094
- Second Approach
 - Avg : 2134.448
 - AE: 1510.557
 - U: 4523.791
 - H: 368.996